

Project Executable Files

Date	02 July 206
Team ID	SWTID-2026-2030
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	3 Marks

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes/ No / NA)
1	Complete source code (all files and folders)	YES
2	README / Setup Guide (instructions to run the project)	YES
3	requirements.txt / package.json / dependency file	YES
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	YES
7	APK / Executable binary (if applicable for mobile/desktop apps)	NO
8	Dockerfile / Containerization config (if applicable)	NO
9	Test files and test results	YES
10	Demo video or walkthrough (if required)	YES

2: File / Folder Structure

Voice-Based-Concept-Understanding-Analyser/

```
| — app.py          # Main Streamlit application
| — speech_to_text.py  # Speech-to-text conversion
| — semantic_eval.py  # Semantic similarity evaluation
| — scoring_engine.py  # Concept scoring logic
| — audio_utils.py    # Audio processing utilities
| — report_generator.py # Report generation
| — pdf_report.py     # PDF report creation
| — requirements.txt  # Python dependencies
| — packages.txt      # System packages
| — tests/
|   | — sample.wav    # Sample audio for testing
| — outputs/
|   | — report.pdf     # Generated PDF report
|   | — waveform.png  # Generated waveform image
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://voice-based-concept-understanding-analyser-11.streamlit.app/
Login Credentials (Demo)	Not Required
Platform / Hosting Provider	Streamlit Cloud
Repository Link	https://github.com/sowmyaanala/Voice-Based-Concept-Understanding-Analyser-Docs.git
Demo Video Link	https://drive.google.com/file/d/1saE4wKLMRPcP2fTpI3Apo6yFjrErrKVf/view?usp=sharing

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites:

Python 3.10 or above

Visual Studio Code

FFmpeg

Git

Internet connection (for Whisper model download)

Steps:

Step 1:

Clone the repository

```
git clone <Repository URL>
```

Step 2:

Open the project folder

Step 3:

Install dependencies

```
pip install -r requirements.txt
```

Step 4:

Run the application

```
streamlit run app.py
```

Step 5:

Open the browser

```
http://localhost:8501
```

Step 6:

Upload an audio (.wav) file.

Step 7:

View the transcript, concept analysis, waveform visualization, and download the generated PDF report.

Step 5: Known Issues / Limitations

S.No	Known Issue	Workaround / Status
1	Works best with clear audio	Use noise-free recordings
2	Supports mainly English speech	Future version will support multiple languages
3	Large audio files take longer to process	Keep audio under 5 minutes